

Quercetagitrin Ameliorates Alzheimer-Like Pathologies Associated with TNF α /NF κ B Pathway in APP/PS1 Mice

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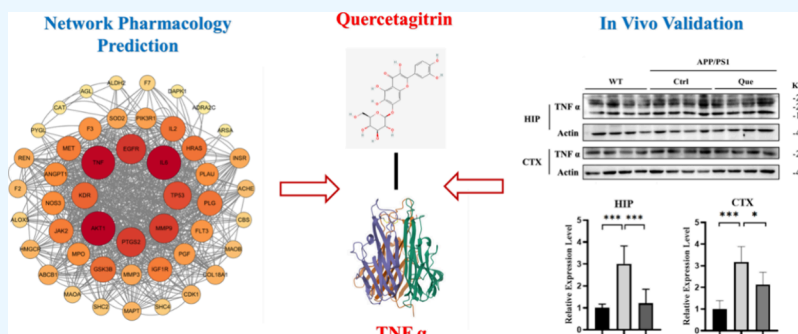
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ABSTRACT: *Background:* Alzheimer's disease (AD) is a neurodegenerative disease with two pathological features in the brain: amyloid β ($A\beta$) plaques and tau tangles. Neuroinflammation plays an important role in the development of AD, closely related to both $A\beta$ and tau pathologies. Tumor necrosis factor α (TNF α) and nuclear factor- κ B (NF κ B) behave as key regulators of neuroinflammation in AD. It is pressing to develop effective AD drugs. *Objective:* This study aimed to explore the effects and mechanisms of quercetagitrin in AD using a combination of network pharmacology analyses and in vivo experiments. *Methods:* The potential target of quercetagitrin in AD was predicted by network pharmacology. The interaction between the compound and the target protein was measured by molecular docking. The in vivo effects were performed in APP/PS1 mice via mouse behavior tests, Western blotting, ThS staining, immunohistochemical staining, and immunofluorescence staining. *Results:* First, network pharmacology analyses were conducted to predict the primary target of the compound, which is TNF α . Then, molecular docking showed that quercetagitrin interacts with TNF α with a high affinity. Finally, the level of TNF α was reduced, and the activation of NF κ B signaling was inhibited by quercetagitrin in APP/PS1 mice. Meanwhile, quercetagitrin treatment ameliorated $A\beta$ pathology, cognitive impairments, and neuroinflammation in the AD mice. *Conclusions:* These findings demonstrate quercetagitrin as a potential therapeutic drug for AD.

1. INTRODUCTION

Alzheimer's disease (AD) is a neurodegenerative disease that predominantly affects older adults. The primary pathological characteristics are the accumulations of amyloid β ($A\beta$) peptide and hyperphosphorylated tau protein, called senile plaques and neurofibrillary tangles, respectively.¹ $A\beta$ plays a key role in the mechanism studies and drug designs for AD.² The progression of the disease is positively related to the increased levels of CSF $A\beta$ peptides³ and brain $A\beta$ plaques.⁴ The amyloid cascade hypothesis proposes that extracellular aggregation of $A\beta$ is the initial step of AD development, followed by tau fibrilization and then neuronal death.⁵ The $A\beta$ peptides are produced from amyloid precursor protein (APP) by the cleavage of the β -site APP cleaving enzyme (BACE1) and γ -secretase. The toxic $A\beta$ aggregates could lead to neuroinflammation, oxidative stress, synaptic impairment, mitochondrial damage, autophagy dysfunction, and so on.⁶

Neuroinflammation involves the activation of microglia and astrocytes responding to pathological alterations, potentially

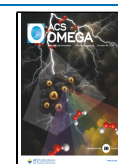
exacerbating neuronal damage. Neuroinflammation has been considered another critical hallmark of AD.⁷ Gene-regulatory network analysis of postmortem brain tissues from AD patients showed that neuroinflammation not only participates in the progress of AD but also is involved in the onset.⁸ Tumor necrosis factor (TNF) plays a critical role in mammalian inflammatory responses. It leads to inflammation either by stimulating inflammatory gene expression or triggering cell death.⁹ Binding of TNF to TNF receptor 1 (TNFR1) promotes inflammation via the activation of the nuclear factor- κ B (NF κ B) and mitogen-activated protein kinase

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(MAPK) signaling pathways, collectively resulting in the transcriptional increase of genes encoding proinflammatory factors.¹⁰ TNF α , the commonest form of TNF, was shown to upregulate the activity of BACE1, as well as NF κ B, which are directly responsible for the production of A β .¹¹

Quercetagitrin (6-hydroxyquercetin 7-O- β -glucopyranoside) is a natural flavonoid compound extracted mainly from marigold (*Tagetes erecta* L.), originally from Mexico and now worldwide. Traditionally, *T. erecta* is utilized as a medicinal plant, tea, cooking ingredient, and food colorant. Quercetagitrin has been shown to exhibit antioxidant activities by either metal chelating or lipoxygenase inhibition.¹² It also could suppress the catalytic activity of PTPN6 and PTPN9, indicating its potential in type II diabetes treatment.¹³ In addition, anti-inflammatory effects of quercetagitrin have been reported. For example, it displayed 60% inhibition of lysosomal enzyme secretion and arachidonic acid release in rat neutrophils.¹⁴ Our laboratory previously presented that it could attenuate NF κ B activation and rescue cognitive deficits by inhibiting tau aggregation in the P301S-tau transgenic mouse model.¹⁵ In this study, we applied quercetagitrin to the APP/PS1 mouse model, which is a very popular model for AD studies. First, network pharmacology was performed to explore the target of quercetagitrin in AD, and TNF was predicted. Next, computer-based molecular docking was employed to observe the interaction of the compound and TNF α . Then, the APP/PS1 mice were administered by quercetagitrin. The in vivo experiments illustrated that the compound could improve cognitive disorders and A β pathologies and alleviate inflammation via the TNF α /NF κ B signaling pathway. Our results provide both computational and experimental evidence for further development of quercetagitrin as a potential drug candidate for AD treatment.

2. MATERIALS AND METHODS

2.1. Network Pharmacology Analysis. A comprehensive search was conducted using the keywords “Alzheimer disease” across several databases, including Gene Cards (<https://www.genecards.org/>), the Therapeutic Target Database (<https://db.idrblab.net/ttd/>), and the DisGeNET database (<https://www.disgenet.org/>). Duplicate entries were subsequently removed. The information of quercetagitrin, including the CAS number and canonical SMILES representation, was searched in the PubChem (<https://pubchem.ncbi.nlm.nih.gov/>) database. This information was then uploaded to the Swiss Target Prediction (<http://swisstargetprediction.ch/>) and Similarity Ensemble Approach databases (<https://sea.bkslab.org/>) to identify the action targets. Additionally, the CAS number was submitted to the TCMSP Database (<https://old.tcmsp-e.com/tcmsp.php>) to retrieve the MOL2 format structure file of quercetagitrin, followed by further analysis in the PharmMapper database (<https://www.lilab-ecust.cn/pharmmapper/>) to obtain additional action targets. Nonhuman gene entries were excluded, and the remaining target names were standardized using UniProt (<https://www.uniprot.org/>). Duplicate targets were eliminated to obtain a comprehensive list of target proteins. The intersection of the quercetagitrin targets with AD targets was calculated to identify the conjunct targets. Finally, both quercetagitrin targets and AD targets were uploaded to the BioLadder platform (<https://www.bioladder.cn/>) for visual analysis, generating a Venn diagram to elucidate the overlapping objectives.

2.2. Construction of the PPI Network. The potential therapeutic targets identified from the network pharmacology analysis were uploaded to the STRING database (<https://string-db.org/>, version 11.0) for further analysis. The species was specified as *Homo sapiens*, and the network type was set to “full STRING network.” The confidence level was adjusted to “medium confidence (0.400),” and the false discovery rate strictness was set to “medium (5%),” while all other parameters were maintained at their default settings to initially obtain the PPI network data. The resulting PPI network was constructed by Cytoscape (version 3.10.2). Topological analysis was conducted using the “Analyze Network” function within Cytoscape. Core targets were identified through screening based on degree, betweenness centrality, and closeness centrality metrics. The degree values were used to visually highlight the targets.

2.3. GO and KEGG Pathway Enrichment Analysis. The potential therapeutic targets were uploaded to the Metascape platform (<http://metascape.org>) for enrichment analysis. The species was designated as *H. sapiens*, and personalized analysis parameters were configured with a minimum overlap of 3, a *P* value cutoff of 0.01, and a minimum enrichment threshold of 1.5. The GO enrichment analysis and KEGG pathway enrichment analysis were performed separately.

2.4. Molecular Docking. The gene names were retrieved from the UniProt database (<https://www.uniprot.org/>) to obtain the corresponding human protein UniProt IDs. These IDs were then utilized to source the crystal structures from the RCSB Protein Data Bank (<https://www.rcsb.org/>), specifying *H. sapiens* as the source organism. Preference was given to crystal structures with low-resolution (Å) values and minimal missing amino acid residues in the active regions relevant to ligand binding. The PDB files of the selected protein structures were downloaded and saved. Additionally, the chemical structure of quercetagitrin was obtained from the TCMSP database and converted to the MOL2 format. Subsequently, the protein structures underwent preprocessing by Chimera software, which involved the removal of water molecules and ligands and the addition of hydrogen atoms. The model/refine loops tool was employed to supplement any missing nonterminal amino acid residues, and the modified structures were saved as PDB files. The information on the proteins used in this study is listed in Table S1. Molecular docking was performed using AutoDock Tools software, including both semiflexible and flexible docking simulations. Visualization of the docking results was accomplished by using PyMOL and LigPlus.

2.5. Animals and Drug Treatment. The APP/PS1 transgenic mice were purchased from Yangzhou Startup Biotechnology Co., Ltd. (Jiangsu, China) and the C57BL/6 mice from Guangdong Medical Laboratory Animal Center. The mice were randomly assigned to the treatment or control groups (10 mice in each group, half male and half female). The mice were housed in groups of 4–5 per cage with ad libitum access to food and water and were maintained under a 12 h light/dark cycle (lights on at 7:00 p.m., off at 7:00 a.m.) at a stable temperature (22 $\pm$ 2 °C). Quercetagitrin (98% purity, HY-N4150, MedChemExpress, Shanghai, China) was first dissolved in pure DMSO and then diluted in drinking water. In the treatment groups, the final quercetagitrin concentration is 10 μ g/mL, and the DMSO content is 0.05% (v/v). The daily uptake of the compound is about 50 μ g as calculated by the average of 5 mL of water intake. The compound concentration

is determined as previously described.¹⁵ The treatment was performed on 6 month-olds to 9 month-olds. Water containing only 0.05% (v/v) DMSO was given to the APP/PS1 mouse-vehicle group and C57 mice. Mice were euthanized after behavioral tests. All animal protocols were approved by the Institutional Animal Care and Use Committee of Shenzhen University Medical School (permit number: IACUC-202300171).

2.6. Open-Field Test. Mice were placed in an open square arena (40 × 40 × 40 cm, length × width × height) and allowed for free movement. The central zone area was defined as 24 × 24 cm. Each mouse was placed in the center area of the arena and permitted for a 5 min free exploration. Their activity was videotaped from the top of the arenas by a video camera. Key metrics included the number of entries into the central open area, the total time spent in the central area, and the overall distance traveled.

2.7. Novel Object Recognition Test. During the adaptation phase, mice were allowed to explore the empty test box for 5 min. Following a 24 h interval, two identical red cylinders were put in the corners of the box for a 5 min training session. After another 24 h, one red cylinder was replaced with a green cuboid, and the mice were allowed to explore for 5 min. Collecting data included the recognition index for new versus old objects and the discrimination index for new objects among different mouse groups.

2.8. Novel Location Recognition Test. During the adaptation phase, mice were positioned in the center of the test chamber and permitted to explore freely for 5 min. After a 24 h delay, a training session commenced with two identical objects placed in opposite corners for another 5 min exploration. One hour later, during the testing phase, the position of one object was altered, and the mice were allowed to explore for 5 min. The time spent exploring the new versus old object locations was recorded, with subsequent statistical analysis conducted on the recognition and discrimination indices.

2.9. Y-Maze Test. Mice were placed in the center of the Y-maze with one arm facing the entrance and allowed to explore freely for 5 min. The exploration sequences in each arm were recorded. The frequency of exploration of three consecutive new arms was considered as an actual alternation. The alternation rate was calculated as follows: actual alternation/(maximal alternation − 2).

2.10. Morris Water Maze Test. Mice were gently placed into one quadrant of the circular pool with a platform facing the wall. The mice were allowed to swim freely to find the platform with a maximum time of 60 s. If the platform was found within 60 s, time was recorded and the mice remained on it for 15 s; if not, they were guided to it and stayed on it for 15 s. This process was repeated across three quadrants over 5 days. The platform was removed on the sixth day. The mice were placed into the opposite quadrant of the one with the platform, and their movements, entries into the target area, and time spent in the target area were recorded and compared. The intelligent V3.0 computer tracking system was used to record the movement of the mice in all of the above behavioral tests.

2.11. Cardiac Perfusion and Brain Tissue Processing. Prior to cardiac perfusion, 4% chloral hydrate (C875281, Macklin, Shanghai, China) was injected to anesthetize the mouse based on the weight of the mouse (0.1 mL/10g). After the limbs were immobilized, an abdominal incision was made to expose the heart. Precooled saline was perfused via a peristaltic pump for 24 min, followed by initial fixation with 4%

paraformaldehyde (G1101, Servicebio, Wuhan, China) for 8 min. Subsequently, the neck was severed to excise the brain tissue, which was soaked in 4% paraformaldehyde for 24 h, dehydrated in 15% sucrose (S818046, Macklin, Shanghai, China), and then treated with 30% sucrose. Frozen sections were prepared after the tissue sank. Brain tissue was washed with PBS (PB180327, Pricella, Wuhan, China), sliced, embedded in the OCT embedding agent (4583, GOLab, Yancheng, China), and stored at −20 °C for solidification. Finally, sections were made with a Leica CM1950 microtome with a thickness of 30 μm and soaked in antifreeze (PBS:ethanol:glycerol 5:2:3), and stored in a −20 °C refrigerator. The specific anatomical regions analyzed in our study included the hippocampal and cortical regions of the mouse brain. The Mouse Brain in Stereotaxic Coordinates (Third Edition) is used as a standard atlas to ensure precise definition and reproducibility.

2.12. Protein Extraction. The mouse brain sample lysate was prepared with the addition of protease and phosphatase inhibitors (78440, Thermo Scientific, Waltham, MA, USA), and RIPA lysate (p0013b, Beyotime, Shanghai, China). A mass ratio of 1:15 (mouse tissue to RIPA lysate) was maintained in the 1.5 mL centrifuge tubes, using one grinding ball for hippocampal tissue and two for cortical tissue. Samples were ground in a freezing grinder at 4 °C (65 Hz, 10 min) and then centrifuged at 4 °C (12,000 rpm, 25 min). The supernatant was transferred into a new tube and sonicated by an ultrasonic crusher (15% power, 2 s sonication, 2 s interval) for 1 min. Protein concentrations were determined using the BCA analysis kit (p0009, Beyotime, Shanghai, China).

2.13. Western Blotting. SDS-PAGE gel (pg112-113, Epizyme, Shanghai, China) was prepared according to the manufacturer's instructions. Protein samples were mixed with 4× loading buffer (2.42 g of Tris, 8 g of SDS, 1.2 mL of concentrated HCl, 40 mL of glycerol in ddH₂O diluted to 100 mL) and 10× 1-bromo-3-mercaptop (bromophenol blue and β-mercaptoethanol at a 1:3 ratio). Samples were denatured by heating at 95 °C for 8–10 min. Protein markers and samples were loaded onto the gel, followed by electrophoresis first at 65 V for 30 min, and then at 130 V until the bromophenol blue reached the bottom of the gel. Postelectrophoresis, proteins were transferred from the gel to a nitrocellulose (NC) membrane. The membrane was sealed with blocking solution (p0252, Beyotime, Shanghai, China) for 30 min, then incubated with the primary antibody at 4 °C overnight. The following primary antibodies were used in this study: beta Actin (66009-1-Ig, Protein Tech, Wuhan, China, 1:5000), Sappα (11088, Immuno-Biological Laboratories, Shanghai, China, 1:1000), Sappβ (10321, Immuno-Biological Laboratories, Shanghai, China, 1:1000), BACE1 (ab108394, Abcam, Beijing, China, 1:1000), PSD95 (36233S, Cell Signaling Technology, Danvers, MA, USA, 1:1000), Synaptophysin (F1215, Selleckchem, Houston, TX, USA, 1:10000), Tubulin (S0B0004, Starter, Hangzhou, China, 1:1000), TNFα (11948S, Cell Signaling Technology, Danvers, MA, USA, 1:1000), iNOS (13120S, Cell Signaling Technology, Danvers, MA, USA, 1:1000), COX-2 (12282S, Cell Signaling Technology, Danvers, MA, USA, 1:1000), NFκB (ab76302, Abcam, Beijing, China, 1:1000), p-NFκB(S468) (3039S, Cell Signaling Technology, Danvers, MA, USA, 1:1000), p-NFκB(S536) (3033S, Cell Signaling Technology, Danvers, MA, USA, 1:1000). After washing to remove unbound primary antibodies, a diluted secondary antibody was added and incubated at

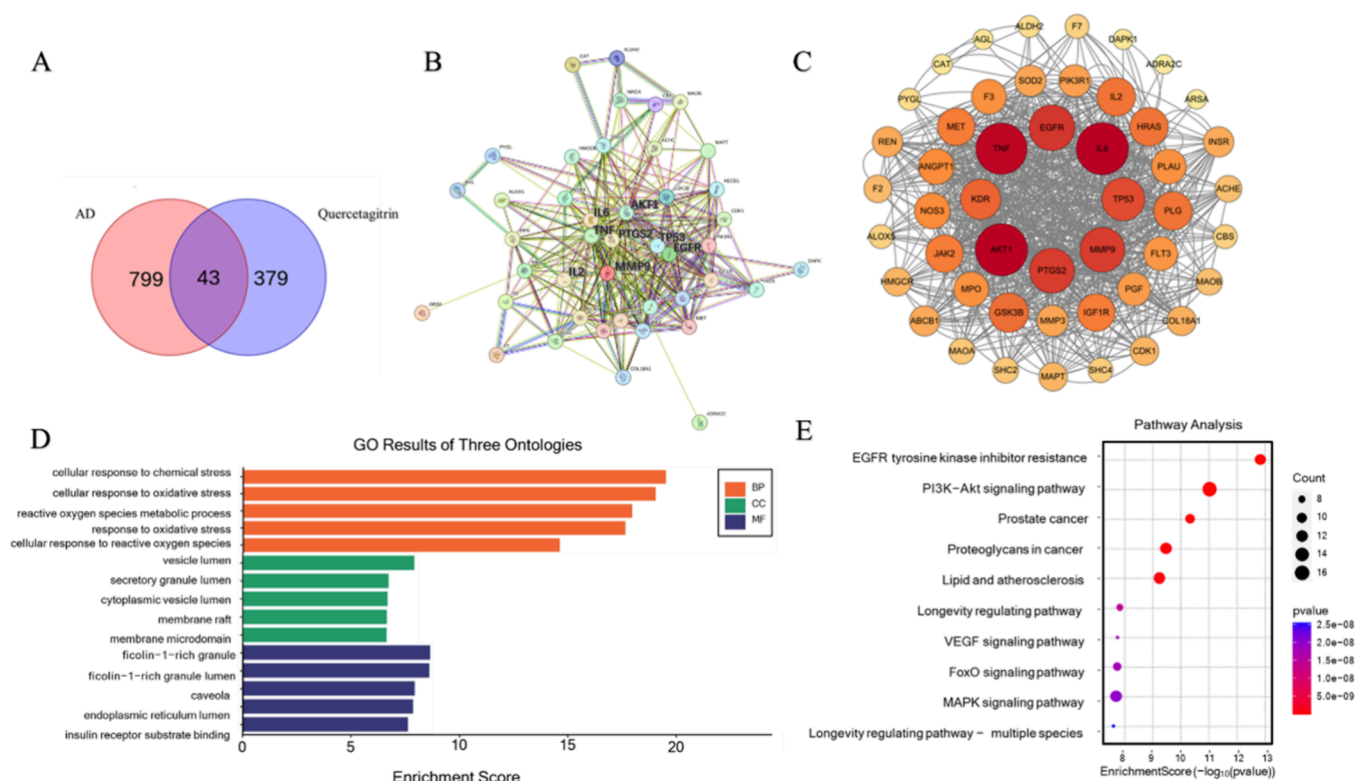


Figure 1. Target prediction of quercetagritin in AD by network pharmacology. (A) Intersection of quercetagritin and AD targets produced by network pharmacology analysis. (B) Quercetagritin-AD intersection target PPI network. (C) PPI network arranged in degrees. (D) Histogram of the top five pathways in each category of GO enrichment analysis. (E) Bubble chart of KEGG pathway enrichment analysis.

room temperature for 60–90 min. Lastly, the NC membrane was washed three times with $1\times$ TBST and visualized using an enhanced chemiluminescence kit (P0018FM, Beyotime, Shanghai, China). Quantification of Western blotting bands was measured with ImageJ software, and the intensity of objective bands was calculated by normalizing the gray value around the strip. The density of the target protein band was normalized to a loading control from the same lane to account for variations in sample loading and transfer efficiency. The result was expressed as a target protein/loading control ratio.

2.14. ThS Staining. The mouse brain slices were put into a 12-well plate and washed with PBS solution three times for 5 min each. Then, 0.3% ThS solution (diluted in 50% ethyl alcohol) was added to the plate. After a 10 min incubation, the brain slices were washed with 50% alcohol twice for 5 min each, transferred to a glass slide, and gently unfolded with a brush. An appropriate amount of a water-based sealing agent was added at the bottom of the slide, and a clean cover glass was placed over the slices. The four corners of the cover glass were sealed with a transparent nail polish. Samples were imaged by a fluorescence microscope (BX53, Olympus, Tokyo, Japan). The quantification was completed by the ImageJ software with default threshold parameters and a “dark background” applied.

2.15. Immunohistochemical Staining. The mouse brain slices were placed in a 12-well plate and subjected to three washes with PBS solution. Endogenous peroxidase blockers were applied to the slices for 10 min, followed by additional washes. The primary antibody 6E10 (803001, BioLegend, Beijing, China) was added to the wells and incubated at $4\text{ }^{\circ}\text{C}$ for 12–16 h. After washing with PBS, the slices were mounted onto slides and incubated with the HRP-conjugated secondary

antibody at room temperature for 20 min, followed by PBS washes in the absence of light. The slices were immersed in DAB color development solution for 8 min. Subsequently, gradient dehydration (50%, 75%, and 95% ethanol) was performed, and the slices were immersed in xylene solution until transparent. Finally, a neutral resin was used to cover the slides. Samples were visualized using a pathological slice scanner (Aperio CS2, LEICA, Wetzlar, Germany). The quantification was completed by the ImageJ software with default threshold parameters and “dark background” applied.

2.16. Immunofluorescence Staining. The mouse brain slices were transferred to a 12-well plate. 4% paraformaldehyde solution was used for immobilization, and then 500 μL of Triton X-100 (p0096, Beyotime, Shanghai, China) was added to permeabilize the cell membranes, incubating at room temperature for 10–15 min. After Triton X-100 was removed, 500 μL of immunostaining blocking solution was applied for 10–15 min. The primary antibody 6E10 was diluted according to the manufacturer’s instructions and incubated at $4\text{ }^{\circ}\text{C}$ for 12–16 h. After washing with PBS for 5 min, the Alexa Fluor488/555 conjugated secondary antibody was mixed at $37\text{ }^{\circ}\text{C}$ for 60 min, avoiding light. Following the removal of the secondary antibody, DAPI (c1002, Beyotime, Shanghai, China) was diluted at 1:500 and added to the slices for 8 min. Finally, the brain slices were transferred to slides, and a water-based sealing agent was applied before covering with a clean cover glass. Once the sealing agent dried, the corners of the cover glass were sealed with a transparent nail polish. Samples were imaged by a fluorescence microscope (BX53, Olympus, Tokyo, Japan). The quantification was completed by the ImageJ software with default threshold parameters and “dark background” applied.

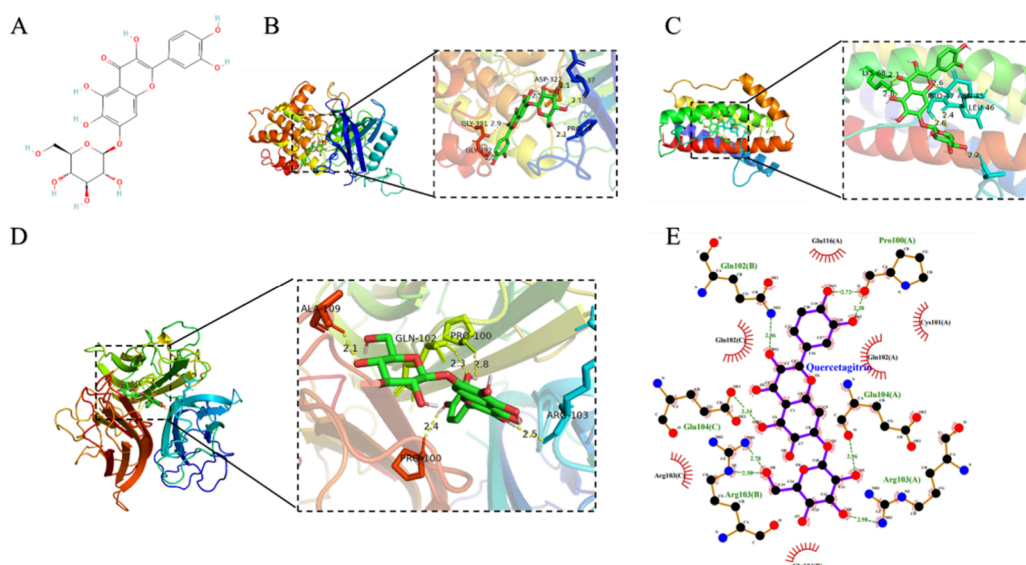


Figure 2. Binding of quercetagitrin and TNF α in molecular docking. (A) Chemical structure of quercetagitrin. 3D-molecular docking diagram of quercetagitrin and AKT1 (B), IL6 (C), and TNF α (D) under semiflexible docking. (E) 2D-molecular docking diagram of quercetagitrin and TNF α under flexible docking.

2.17. ThT Fluorescence Assay. ThT is a cationic benzothiazole dye that exhibits enhanced fluorescence when bound to amyloid fibrils. The A β_{1-42} peptide was purchased from ChinaPeptides CO., Ltd. (Shanghai, China). Five μ M ThT, 40 μ M A β peptide, and different concentrations of quercetagitrin (0, 20, and 40 μ M) were mixed in PBS buffer in a 96-well plate and incubated at 37 °C for 48 h. The final fluorescence values were measured by a microplate reader (Synergy H1, Biotek, Winooski, VT, USA) with an excitation at 440 nm and emission at 485 nm. Each experiment was biologically repeated five times.

2.18. Transmission Electron Microscopy (TEM). The effect of quercetagitrin on the morphologies of the A β fibrilization was investigated by TEM imaging. In brief, the protein samples from the end of the ThT fluorescence experiments were diluted 10-fold with ddH₂O and spotted onto a 230-mesh carbon-coated copper grid (BZ1102, EMCN, Beijing, China). After incubation for 30 min, the residual solution was removed with filter paper. The grids were washed with ddH₂O and mixed with 1% uranyl acetate for 30 s. The morphology of A β fibrils was observed using TEM (JEM1230, Tokyo, Japan).

2.19. Statistics Analysis. Results are shown as mean $\pm$ SEM. Statistical analysis was performed using GraphPad Prism 8.0.2 statistical software. Statistical significance was assessed using Student's *t* test, one-way or two-way ANOVA. *P* values of 0.05 or less were considered to denote the significance. The confidence interval is 95%. The effect sizes *d* is calculated by $d = \frac{M1 - M2}{SD_{pooled}}$. *M1* and *M2* are the means of the two groups, and *SD_{pooled}* is the combined standard deviation of the two groups.

3. RESULTS

3.1. Network Pharmacology Predicts the Targets of Quercetagitrin in AD. Network pharmacology is a burgeoning approach for Chinese herbal medicine research, especially in drug discovery.¹⁶ In our network pharmacology analysis, a total of 842 and 422 targets related to AD and quercetagitrin were identified across various databases, yielding

43 overlaps (Figure 1A). The full genes in each group are listed in Table S2. The relationships among these targets were illustrated in Figure 1B. 43 nodes and 303 edges were revealed, resulting in an average degree of 14.1 and an average local clustering coefficient of 0.731. The enhanced protein–protein interaction (PPI) network visually emphasizes targets based on their degree values, transitioning from light yellow to dark red and from smaller to larger areas. Topological analysis of the PPI network nodes yielded average values of degree centrality, betweenness centrality, and closeness centrality of 15.375, 0.4279, and 0.5730, respectively. The top three predictions were for IL-6, AKT1, and TNF (Figure 1C).

Furthermore, Gene Ontology (GO) enrichment analysis resulted in 3083 entries, with 2688 related to Biological Processes (BP), 153 related to Molecular Functions (MFs), and 242 related to Cellular Components (CCs). The identified potential therapeutic targets were primarily localized in the vesicle lumen, secretory granule cavity, and cytoplasmic capsule. These targets play critical roles in regulating substrate binding of the insulin receptor and the activity of transmembrane receptor protein tyrosine kinases, impacting various biological processes such as cellular responses to chemical stress, oxidative stress, and reactive oxygen species metabolism (Figures 1D and S2). Additionally, the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis revealed a total of 262 pathways. The top ten pathways were selected based on $-\log_{10}$ (*P* value) and represented in a bubble chart (Figure 1E). In this chart, larger bubbles correspond to a greater number of genes associated with the intersecting targets within the pathway, while the higher intensity of the red color indicates a smaller *P* value, reflecting greater significance in enrichment. Notably, our findings suggest that quercetagitrin might modulate AD through mechanisms involving resistance to EGFR tyrosine kinase inhibitors, the PI3K/Akt signaling pathway, prostate cancer pathways, and lipid and atherosclerosis.

3.2. Molecular Docking Reveals the Interaction of Quercetagitrin and TNF α . Computer-based molecular docking was then carried out to observe the interaction of

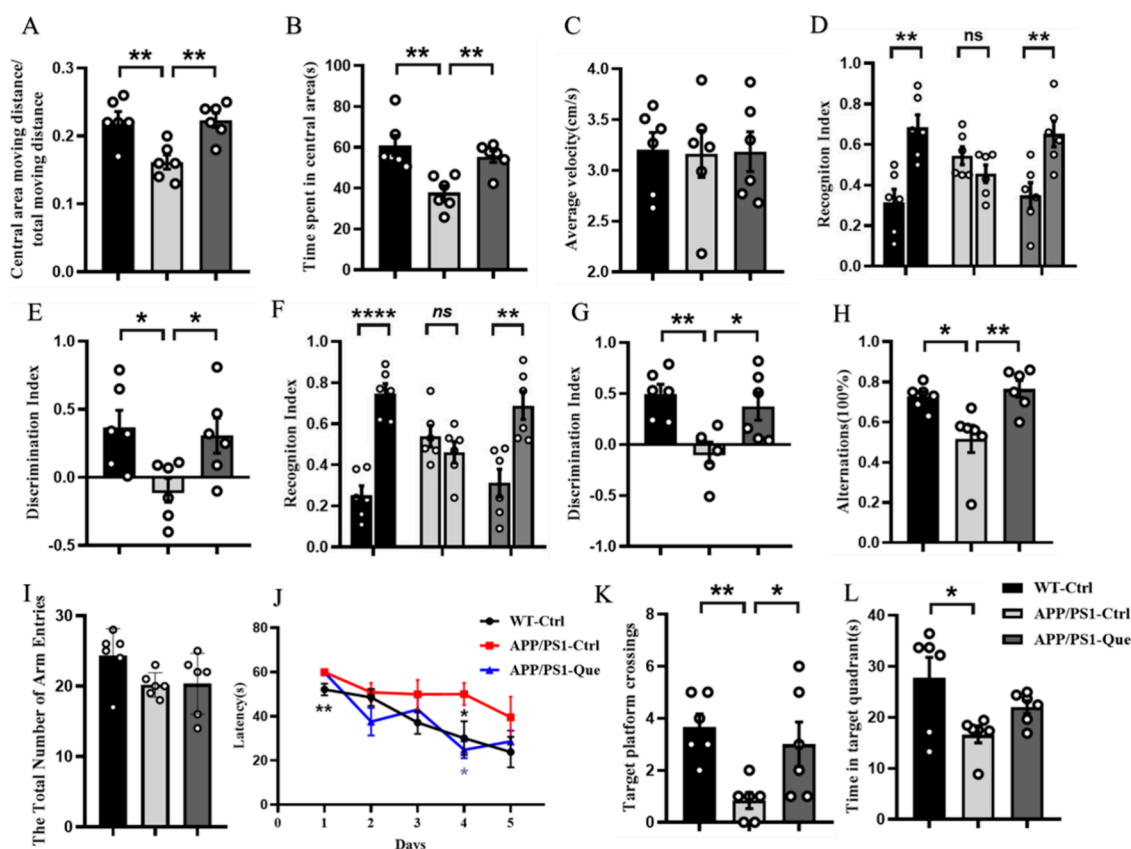


Figure 3. Behavioral experiments showed improvement of anxiety and cognition in APP/PS1 mice by quercetagritin. The movement distance ratio (A) and the residence time (B) of the mice in the central open area in the open-field test. (C) Average speed of mice in each group in the open-field test. (D) Proportion of old (left column in each group) and new objects (right column in each group) exploration of the mice in the novel object recognition experiment. (E) Calculated discrimination index of the mice in the novel object recognition experiment. (F) Proportion of old position (left column in each group) and new position (right column in each group) exploration of the mice in the novel position recognition experiment. (G) Calculated discrimination index of the mice in the novel position recognition experiment. Spontaneous alternation rate (H) and total number of arm entries (I) of the mice in the Y-maze test. (J) Learning curve of the mice in each group during directional guidance training in the Morris water maze test. Platform crossing of the target position (K) and the residence time in the quadrant of the platform (L) of the mice during the detection period in the Morris water maze test. Data were expressed as mean $\pm$ SEM, $n = 6$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.0001$.

quercetagritin with AKT1, IL6, and TNF α predicted by the network pharmacology analysis. Figure 2A displays the structure of quercetagritin used in the molecular docking. The docking results indicated that quercetagritin could directly bind to AKT1, IL6, and TNF α , with the binding energy of -8.5 , -7.3 , and -8.1 kcal/mol, respectively, using the semiflexible docking method (Figure 2B–D). However, the level of AKT1 and IL6 did not show any discrepancy between the APP/PS1 mice with or without quercetagritin treatment (Figure S2). Thus, we focused on TNF α in the following studies. The binding energy of -9.41 kcal/mol was given by another docking mode, flexible docking (Figure 2E). It is widely accepted that binding energies of ≤ -5.0 kcal/mol are indicative of favorable interactions between the compound and the protein target, while values of < -7 kcal/mol suggest a strong binding affinity. Our results from both docking approaches demonstrate a putative binding between quercetagritin and TNF α .

3.3. Quercetagritin Improves Anxiety-Like Behaviors and Cognitive Impairment in APP/PS1 Mice. Based on the above computational results, we next explored the effects of quercetagritin in APP/PS1 mice. After administering quercetagritin for 3 months (from 6 to 9 months of age), the mice underwent behavioral tests, including the open-field test, novel

object recognition, novel location recognition, Y-maze, and Morris water maze tests to assess anxiety, learning, and memory. In the open-field test, APP/PS1 mice exhibited pronounced anxiety-like behavior compared to C57 mice, with a significantly reduced residence time and movement distance in the central open area. Quercetagritin treatment markedly improved these anxiety-like behaviors, increasing both residence time and movement distance in the central open area (Figure 3A,B). These differences were not due to variations in movement, as the average move speed showed no significant change among the three groups (Figure 3C). The results suggest that quercetagritin can ameliorate anxiety-like behavior in 9 month-old APP/PS1 mice.

In the novel object recognition experiment, the APP/PS1 mice could not distinguish the old and new objects and even spent more time exploring the old objects. In contrast, both C57 mice and the treated APP/PS1 mice exhibited a significantly higher recognition index in the new objects, suggesting that quercetagritin treatment enhanced exploratory behavior and cognitive function in APP/PS1 mice (Figure 3D,E). Similarly, the novel position recognition experiment gave the corresponding observation and conclusion (Figure 3F,G). In the Y-maze test, APP/PS1 mice exhibited a significantly lower spontaneous alternation rate compared

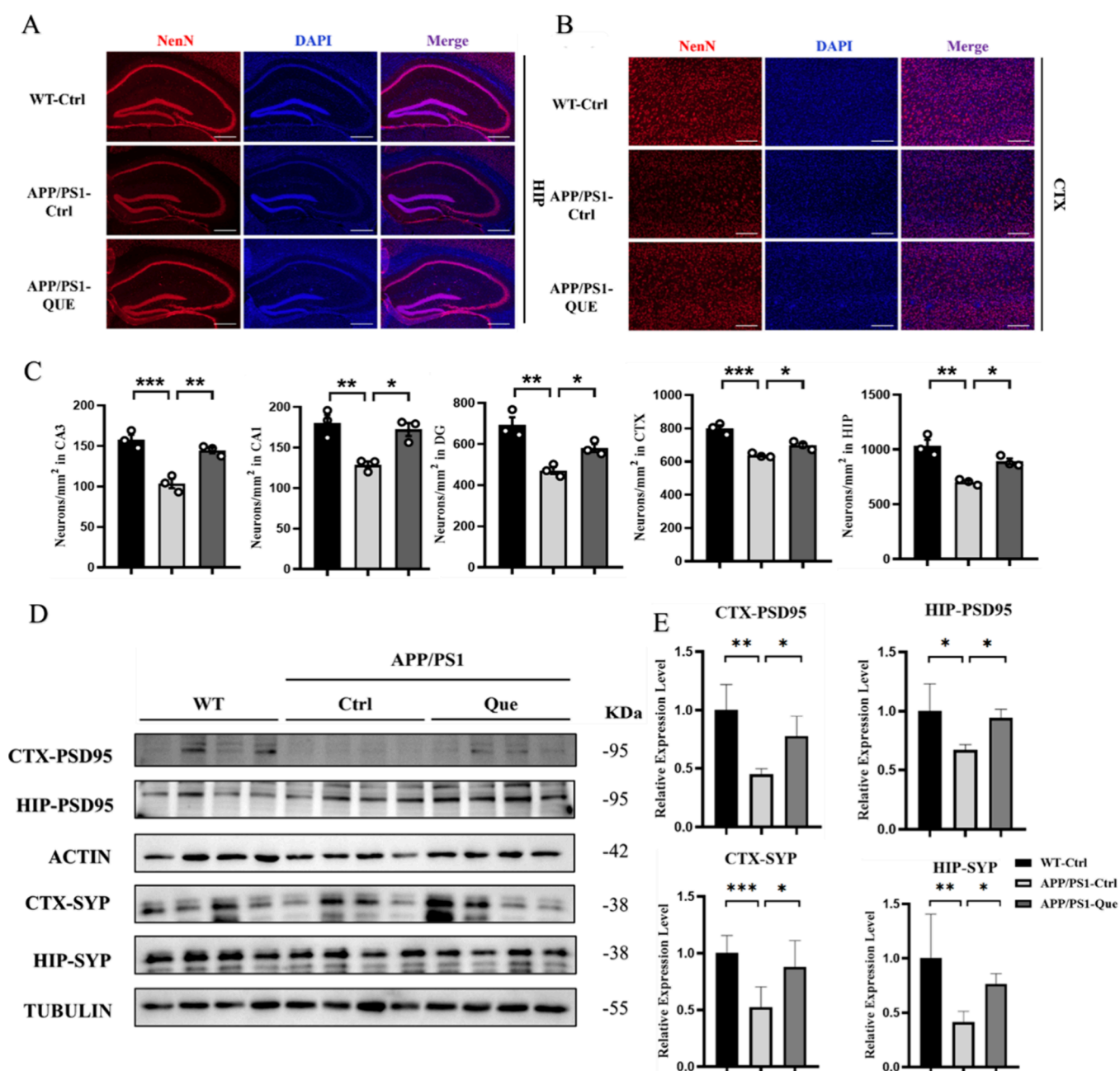


Figure 4. Quercetagritin protects neurons in 9 month-old APP/PS1 mice. The representative immunofluorescence images of NeuN in the hippocampus (A) and the cortex (B) of the mice in each group. Scale bar: 200 μ m. (C) Quantitative analysis of the number of neurons in hippocampal CA1, CA3, DG, and total hippocampus and cortex, $n = 3$. (D) Western blot analysis of PSD95 and SYP in the cortex and hippocampus of the mice in each group. (E) Quantification analysis of the two synaptic proteins, $n = 4$. Data were expressed as mean $\pm$ SEM, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

with C57 mice, while the administration group showed a marked improvement (Figure 3H). The total number of arm entries among the three groups was not significant (Figure 3I). Finally, the Morris water maze test was employed to assess the learning and memory abilities of the mice. During the 5-day training, C57 mice and the treated APP/PS1 mice exhibited superior learning abilities compared with control APP/PS1 mice (Figure 3J). After a 24 h retention period, C57 and the administration mice showed more crossing and residence time in the target platform location compared to control APP/PS1 mice (Figure 3K,L). The results all express that treatment of quercetagritin effectively improves the cognitive dysfunction in 9 month-old APP/PS1 mice.

3.4. Quercetagritin Protects Neuronal Loss in APP/PS1 Mice. Neuronal loss is a critical pathological feature in AD and serves as the cellular basis for cognitive deficits. In this study, the number of neurons was measured via immunofluorescence staining by the neuronal marker NeuN. As shown in Figure 4, compared to C57 mice, APP/PS1 mice exhibited a significant reduction of neuron numbers in all of the CA1, CA3, and DG regions of the hippocampus and in the cortex. However, the administration of mice displayed a significant increase in neuron counts in all these brain regions compared to the control APP/PS1 mice, indicating that quercetagritin treatment effectively restored the neuronal loss in both the hippocampus and cortex.

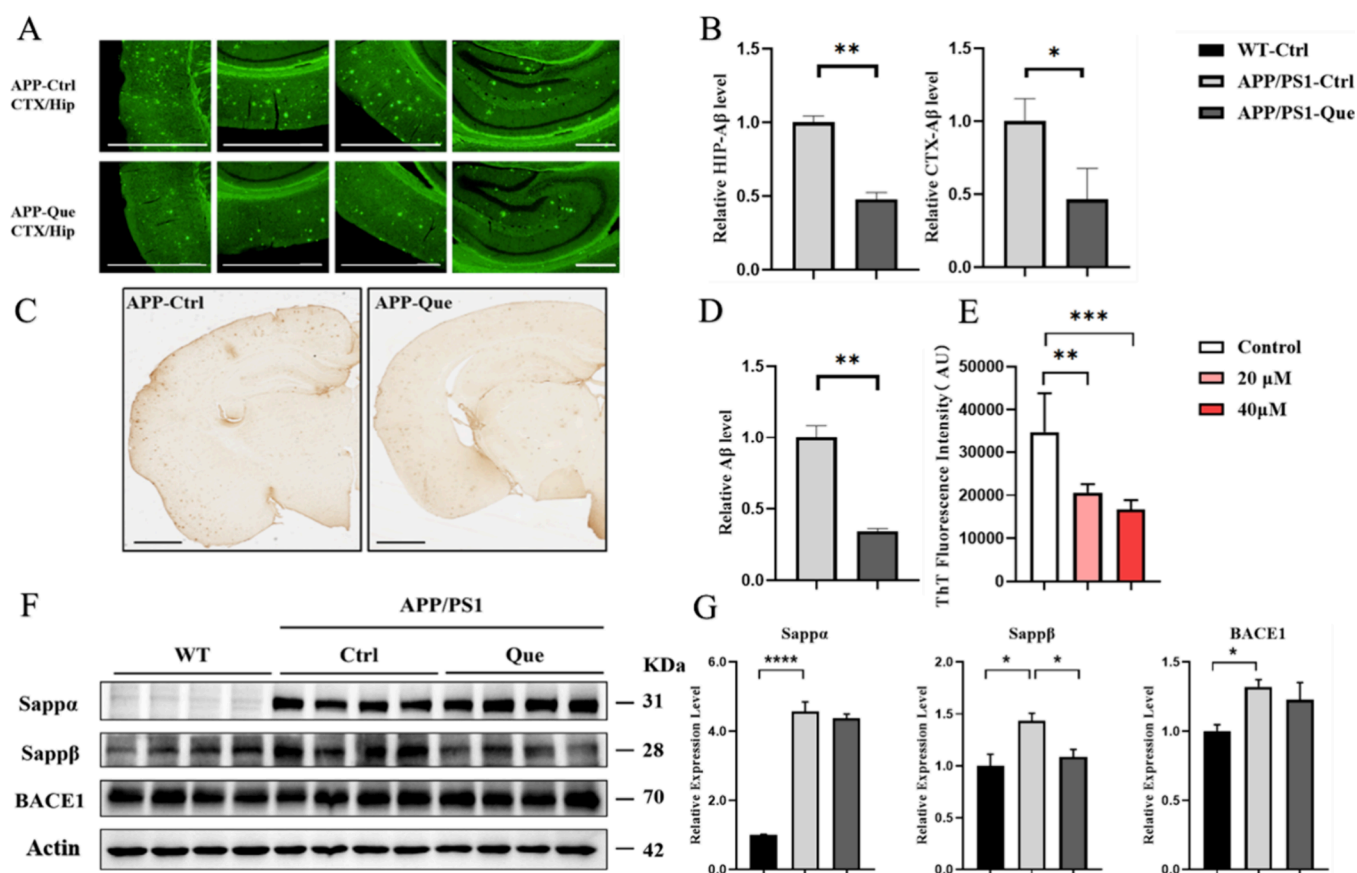


Figure 5. Quercetagitritin alleviates A β pathology in 9 month-old APP/PS1 mice. (A) Representative images of ThS staining of the brain tissues from APP/PS1 mice with or without quercetagitritin. Scale bar: 200 μ m. (B) Quantitative statistical analysis of the ThS intensities in cortex and hippocampus, $n = 3$. (C) Representative images of immunohistochemical staining of the brain tissues from APP/PS1 mice with or without quercetagitritin. Scale bar: 200 μ m. (D) Quantitative statistical analysis of the levels of A β aggregates, $n = 3$. (E) The final fluorescence intensity of ThT in the in vitro A β aggregation solution without or with 20 μ M, 40 μ M quercetagitritin, $n = 5$. (F) Western blot analysis of Sapp α , Sapp β , and BACE1 in the brain of the mice in each group. (G) Quantification analysis of the three proteins, $n = 4$. Data were expressed as mean $\pm$ SEM, * $p < 0.05$, ** $p < 0.01$, **** $p < 0.0001$.

Furthermore, the levels of two synaptic proteins, synaptophysin (SYN) and postsynaptic density protein-95 (PSD-95), were measured in both cortical and hippocampal samples. Synaptophysin is a presynaptic vesicle membrane protein involved in the localized retrieval and recycling of synaptic vesicles after exocytosis. PSD-95 is predominantly or exclusively associated with postsynaptic densities. Both proteins are critical for the maintenance of neurotransmission. The levels of the two proteins in both the cortex and hippocampus decreased in the APP/PS1 mice compared to the WT mice but were restored by the treatment of quercetagitritin (Figure 4D,E). So, quercetagitritin could rescue the synaptic functions of the neurons in the brain of the APP/PS1 mice. The immunofluorescence and Western blotting results are consistent with those revealed by the behavioral tests.

3.5. Quercetagitritin Alleviates A β -Related Pathology in APP/PS1 Mice. To determine whether quercetagitritin treatment can reduce A β pathology in APP/PS1 mice, key indicators were assessed using thioflavin S (ThS) staining, immunohistochemistry, and Western blotting. ThS is usually utilized to detect amyloid plaques that possess abundant β -strand structures. ThS staining experiments indicated a significant reduction in fluorescent plaques in both the cortex and hippocampus of quercetagitritin-treated mice compared to control APP/PS1 mice (Figure 5A,B). Then, immunohisto-

chemical staining using the A β antibody was performed to further observe the A β aggregates. The results also revealed decreased levels of A β plaques in the brain of APP/PS1 mice with quercetagitritin, particularly in the cortex (Figure 5C,D). BACE1 and sAPP β only appear in the amyloidogenic pathway, while sAPP α appears in the nonamyloidogenic pathway. Western blotting analysis of the three proteins is shown in Figure 5F,G. As expected, the levels of sAPP α were indistinctive, but sAPP β decreased significantly in the APP/PS1 mice after quercetagitritin treatment. The levels of BACE1 showed a downward trend without statistical significance. In summary, quercetagitritin improves A β -related pathology in 9 month-old APP/PS1 mice.

Additionally, ThT fluorescence analysis was performed to evaluate the inhibitory effect of quercetagitritin on A β aggregation in vitro. The fluorescent dye ThT exhibits enhanced fluorescence when bound to amyloid fibers, making it commonly used for detecting the aggregation process of amyloid proteins. After a 48 h incubation, the endpoint fluorescence values were measured. As shown in Figure 5E, compared to the control group without treatment, quercetagitritin significantly decreased the ThT fluorescence intensity of the in vitro A β aggregation solution in concentration dependence. Then, TEM imaging was carried out to confirm the fluorescence results. The mature A β fibrils in the control

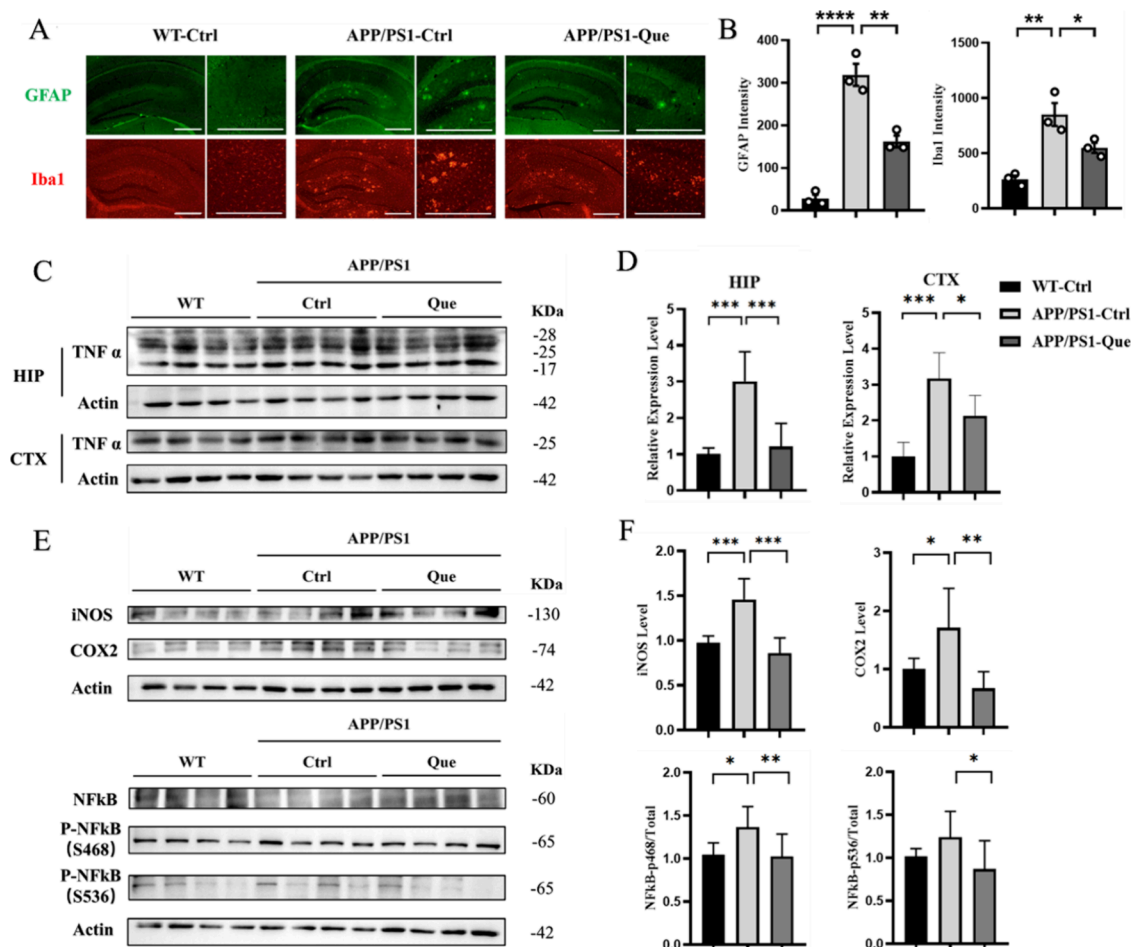


Figure 6. Quercetagitritin improves neuroinflammation in 9-month-old APP/PS1 mice. (A) Representative immunofluorescence images of GFAP and Iba1 in the brain tissue of mice in each group. Scale bar: 200 μ m. (B) Quantitative analysis of the fluorescence intensity of GFAP and Iba1 in the hippocampus. $n = 3$. (C) Western blot analysis of the TNF α in the hippocampus and cortex of mice in each group. (D) Quantification of the level of TNF α . $n = 5$. (E) Western blot analysis of iNOS, COX-2, NF κ B, p-NF κ B(S468), and p-NF κ B(S536) in the hippocampus of mice in each group. (F) Quantification of the levels of the proteins in (E), $n = 4$. Data were expressed as mean $\pm$ SEM, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

group could be described as long and compact fibrils, in contrast to those in the quercetagitritin-treated groups. Quercetagitritin inhibited A β aggregation in a dose-dependent manner, as shown by the reduced A β fibrillization (Figure S3). The results indicate that quercetagitritin could directly inhibit A β aggregation in vitro.

3.6. Quercetagitritin Reduces Neuroinflammation in APP/PS1 Mice. Based on the results of network pharmacology, TNF is one of the most likely therapeutic targets (Figure 1C). TNF is a proinflammatory cytokine that participates in the neuroinflammatory response in vivo. So, we explored whether quercetagitritin could mitigate neuroinflammation via the TNF α signaling pathway in APP/PS1 mice. Astrocytes and microglia are the two primary immune cells in the brain. Immunofluorescence staining revealed significantly elevated levels of astrocyte marker GFAP and microglia marker Iba1 in the hippocampus of APP/PS1 mice compared with C57 mice. Notably, the fluorescence intensity of the two markers was markedly reduced upon the treatment of quercetagitritin (Figure 6A,B). Subsequently, Western blot analysis demonstrated a significant reduction in TNF α protein levels in both the hippocampus and cortex of APP/PS1 mice treated with quercetagitritin (Figure 6C,D). TNF can indirectly activate

NF κ B, leading to the formation of phosphorylated NF κ B (p-NF κ B). Finally, Western blotting was performed to detect the levels of total NF κ B, p-NF κ B, and two downstream inflammatory factors: nitric oxide synthase (iNOS) and cyclooxygenase-2 (COX-2). Expectedly, a decrease in both p-NF κ B (S468)/total NF κ B and p-NF κ B (S536)/total NF κ B was observed in the treatment group. The levels of iNOS and COX-2 were reduced as well upon quercetagitritin (Figure 6E,F). Collectively, these results suggest that quercetagitritin effectively mitigates neuroinflammation through the TNF α /NF κ B pathway in APP/PS1 mice.

4. DISCUSSION

Network pharmacology was first proposed in year 2007, and it is an approach combining chemogenomics with network biology for drug discovery.¹⁷ It highlights the comprehensive effects of disease or syndrome biological networks as targets to clarify the general regulatory mechanism of herbal constituents.¹⁸ Recently, this technology has been used a lot to elucidate the application of traditional medicine in various kinds of diseases, including AD. For example, polymethoxylated flavones, the key lipid-soluble components in *Citri Reticulatae Pericarpium*, have been predicted and validated to

inhibit neuroinflammation in AD mouse models.¹⁹ Berberine ameliorates AD pathological features by inhibiting the JNK-P38MAPK pathway and ferroptosis, further reducing A β plaque deposition and inhibiting inflammatory response in 5 $\times$ FAD mice.²⁰ Aucubin alleviates cognitive and memory deficits, AD-like pathologies of the mouse brain, including the deposition of A β plaques and neuronal damage, and inflammatory responses by inhibiting the ERK/FOS axis in APP/PS1 mice.²¹ The combination of network pharmacology and experiments revealed that steamed *Panax notoginseng* improved the olfactory memory ability, climbing ability, and A β ₄₂ content in the brain of AD *Drosophila* models and significantly reduced the expression of TNF and Toll-like receptor after treatment.²² In this study, network pharmacology analysis indicated TNF α as a major potential target of quercetagitrin in AD. Then, molecular docking suggested that quercetagitrin could directly interact with TNF α with a robust binding affinity. Next, experiments in APP/PS1 mice were carried out to validate the network prediction. Indeed, after treatment with quercetagitrin, the reduced TNF α level as well as the deactivation of NF κ B were observed in the hippocampus of the mice.

TNF is a crucial cytokine that plays a significant role in the regulation of immune and inflammatory responses. Here, the KEGG pathway analysis of the potential therapeutic targets predicted from network pharmacology represented two primary pathways: EGFR tyrosine kinase inhibitor resistance and the PI3K/Akt signaling pathway. A close relationship between TNF and these two pathways has also been reported. Activation of EGFR tyrosine phosphorylation was TNF concentration dependent, and TNF cytotoxic resistance was reversed by the tyrosine kinase inhibitor tryphostin in ME-180R cells.²³ TNF stimulated EGFR transactivation via the TNF α -converting enzyme, and inhibition of EGFR tyrosine kinase activity increased TNF-induced lung epithelial cell apoptosis.²⁴ The knockout of TNF α led to the alteration of miRNA expressions involved in PI3K/Akt signaling in CML K562 cells.²⁵ TNF α also induced the activation of the PI3K/AKT/mTOR pathway in HaCaT and NHEK cells, and blocking this pathway inhibited TNF α -induced inflammation.²⁶ TNF α is one of the major cytokines found in AD brains. TNF α could activate NF κ B signaling in microglia²⁷ and neurons.²⁸ NF κ B is a key regulator of inflammation in the central nervous system.²⁹ Interleukin-17A (IL-17A) generated the microglial secretion of TNF α through the NF κ B signaling pathway, promoting the progression of AD in the APP/PS1 mice.³⁰ The increase of TNF α and activation of NF κ B were again observed in APP/PS1 mice and A β _{25–35} + lipopolysaccharides (LPS)-challenged BV2 microglia.³¹ The activation of TNF α /NF κ B was also reported in LPS-exposed and HDAC7-overexpressed astrocytes.³² In this study, quercetagitrin suppressed TNF α /NF κ B-mediated neuroinflammation in APP/PS1 mice, possibly via directly binding to TNF α . Further in vitro experiments like surface plasmon resonance, biolayer interferometry, and isothermal titration calorimetry could be employed to measure the interaction of TNF α and quercetagitrin, as well as their binding mode and affinity. The inhibition of TNF α /NF κ B-mediated neuroinflammation could restore neuronal function that leads to behavioral improvement of the mice. A Poland research group reported the existence of quercetagitrin in *Xerolekia speciosissima* (L.) Anderb., a plant from Italy. 7,10-diisobutyryloxy-8,9-epoxythymyl isobutyrate, an extraction from the

above plant, significantly reduced TNF α , IL-8, IL-1 β and CCL2 excretion by LPS-stimulated human neutrophils.³³ However, the authors did not present any results on quercetagitrin. Quercetagetin, a structurally similar compound to quercetagitrin, effectively blocked IFN γ and TNF α -induced chemokine CCL17 and CCL22 expressions in HaCaT human keratinocytes.³⁴

Quercetagitrin is a polyphenol flavonoid molecule extracted mainly from *Tagetes erecta* L. The polyphenol-enriched extracts from *T. erecta* L. inhibited glycogen synthase kinase-3 β (GSK-3 β) activity and promoted β -catenin nuclear translocation in MC3T3-E1 cells.³⁵ Actually, GSK-3 β also behaves as the primary kinase on tau hyperphosphorylation in AD. Besides A β plaques, neurofibrillary tangles consisting of hyperphosphorylated tau are another clinical pathology in the brain of AD patients. Our previous article showed that quercetagitrin alone could reduce tau phosphorylation at different residues via GSK-3 β inhibition in PS19 mice and tau-overexpressed HEK293 and N2a cells.¹⁵ One fraction (containing over 85% quercetagitrin) from *T. erecta* L. displayed antioxidant activities, which are metal-chelating activity and lipoxygenase inhibition.¹² The balance of metal ions is essential for human health, especially in the central nervous system, while metal ions participate in the development of AD.³⁶ The levels of Zn²⁺, Cu²⁺ ions, and Fe²⁺ ions are all elevated in the AD brain. So, the metal-chelating activity of quercetagitrin could be beneficial to AD treatment from another perspective. 5-lipoxygenase (5-LOX) is a proinflammatory enzyme that mediates inflammation, and abnormal activation of 5-LOX has been shown in the development of age-related brain pathology, suggesting the inhibition of 5-LOX activation as a promising therapeutic direction for AD.³⁷ The lipoxygenase inhibition function of quercetagitrin plays a positive role in AD treatment. In addition, our current results showed that quercetagitrin reduced A β plaques in the mice, possibly due to the direct inhibition of A β aggregation as it did in vitro. However, more experiments should be carried out to elaborate on this point. AD is a complicated neurodegenerative disease with various inducers, and the current drug acting on a one-single target cannot prevent or cure the disease very well. Thus, the utilization of multiple interventions against several molecular targets of AD pathologies might be the most practical approach to curing the disease.³⁸ Considering the various bioactive functions summarized above, quercetagitrin might be a potential leading compound for the treatment of AD.

In conclusion, quercetagitrin could improve the cognitive deficits, reduce A β levels, and ameliorate neuroinflammation, likely implicating the TNF α /NF κ B pathway in APP/PS1 mice.

■ ASSOCIATED CONTENT

Data Availability Statement

The data are available throughout the manuscript and supporting files.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.5c05682>.

The structure parameters of the proteins used in docking (Table S1); full list of genes from network pharmacology (Table S2); complete pathways from GO enrichment analysis (Figure S1); the levels of AKT1 and IL6 in APP/PS1 mice (Figure S2); TEM images of A β aggregations (Figure S3) (PDF)

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Author Contributions

[#]C.Y. and S.L. contributed equally to the work. C.Y.: Writing—original draft, Methodology. S.L.: Methodology, Data Curation. Y.H.: Software, Data Curation. Y.L.: Methodology. Z.X.: Methodology. G.L.: Software. S.X.: Writing—review and editing, Supervision, Resources, Project administration, Funding acquisition.

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Notes

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ABBREVIATIONS

AD, Alzheimer's disease; A β , amyloid β ; APP, amyloid precursor protein; BACE1, β -site APP cleaving enzyme; TNF, tumor necrosis factor; TNFR1, TNF receptor 1; NF κ B, nuclear factor- κ B; MAPK, mitogen-activated protein kinase; PPI, protein–protein interaction; GO, gene ontology; BP, biological process; MF, molecular function; CC, cellular component; KEGG, Kyoto encyclopedia of genes and genomes; SYP, synaptophysin; PSD-95, postsynaptic density protein-95; TEM, transmission electron microscopy; ThS, thioflavin S; ThT, thio-

flavin T; iNOS, nitric oxide synthase; COX-2, cyclooxygenase-2; GSK-3 β , glycogen synthase kinase-3 β ; 5-LOX, 5-lipoxygenase; NC, nitrocellulose

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